EE/CE 3201 Fundamental Lab I
Electrical and Computer Engineering
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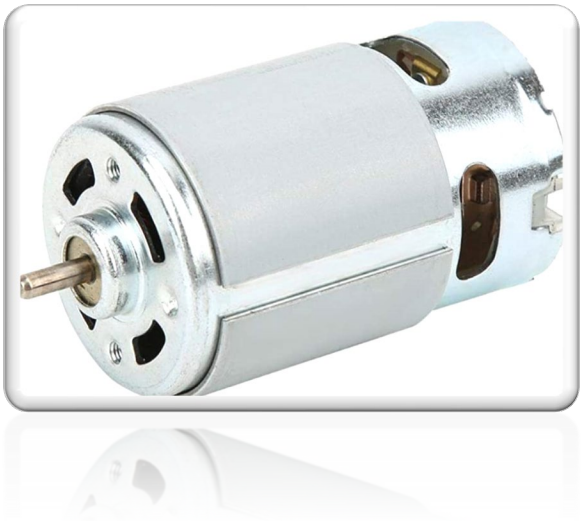

LAB 6:
DC Electric Motors

EE/CE 3201 Fundamental Lab I

Lab #6 – DC Electric Motors

1. Lab Objectives

- Operational Level Understanding of DC electric motors.
- Ability to design an experiment to measure the torque vs speed curve.
- The first part of the lab involves measuring the "No-Load" rotational speed of a P711Ahobby-motor as a function of the input voltage (and electrical power) using a tachometer.
- The second part of the lab involves measuring the torque vs. speed curve of the motor for two different input voltages.
- Finally, you will analyze the torquevs. speed curve to find the maximum mechanical power output of the motor at the two inputvoltages as well as the efficiency curves and stall torques for the motor at those voltages.

2. Introduction

Electric motors convert electrical power into mechanical power either as a rotational motion or a linear motion. You will make measurements on a very common electric motor: thedc electric motor. It converts dc electrical power ($P_{Electrical} = I*V$) into a torque at a rotational speed ($P_{Mechanical} = \tau*\omega$). In this lab, you will make measurements on both $P_{Electrical}$ and $P_{Mechanical}$so that you can learn about the basic operation of electric motors.

Electric motors operate at different rotational speeds depending upon the load placed on them. Heavy loads cause the motor shaft to rotate slower and light loads allow the shaft to rotate faster. The amount of load is described by the amount of torque on the motor's shaft and leads to a torque vs. rotational speed curve. An example is shown in Fig.2.1. When the torque goes to zero, the motor reaches its peak speed called the "No-load" speed. If the torque becomes too large, the motor is unable to spin. The smallest torque which can prevent the motor from rotating is called the "Stall" torque.

In this lab, you will learn about dc electric motors, measure some key operating parameters, analyze those measurements to design a second – more in depth – test on motor torque versus motor speed and then analyze the performance of the motor.

Electric motors convert electrical power ($I*V = Current*Voltage$) into mechanical power either as a rotational motion ($c*\omega = Torque*Angular\ Rotation\ Frequency$) or as a linear motion ($F*v = Force*Velocity$). They impact our lives every day and come in all sizes from microscopic(cell phone vibrators) to gigantic (ship and train propulsion), from very low power (milliwatt andeven microwatt) to very high power (100-megawatt) motors. They enable myriad convenient devices including fans, electric shavers, elevators and escalators, air compressors, drills, sewing machines, vacuum cleaners, computer hard drives, pumps, garage door openers, engine starters and power windows, mixers and food processors as well as treadmills (and other exercise equipment) to name just a few. In addition, they are the basis for all movement in robotics which enables all automation in manufacturing as well as electric vehicles.

Electric motors get sorted into a few basic types based (for the most part) on [1] the power source (DC or AC and single-, two- or three-phase), [2] their internal construction details (brushed or brushless including "stepper" motors as well as permanent magnets or windings) [3] their application and [4] their type of motion (rotating or linear). Despite these different classifications, there is a basic underlying principle for the vast majority of electric motors: They operate using the fact that opposite poles of a magnet attract (and vice versa). Thus, a force can be induced on any coil that is carrying current by another coil carrying current (or by a permanent magnet).

The goal of this experiment is to give you an introduction to electric motors and their operation. You will learn about the maximum amount of torque (rotational "force") that a motor can produce and the maximum speed at which the motor shaft can spin. In addition, youwill learn a little about motor efficiency and the amount of mechanical power it can produce from the input electrical power. You will use the same motors again in Fundamentals 2 when you learn about PID controllers (Systems and Controls class – 4310) because a motor run backwards is an electricity generator.

Background

To describe electric motor operation one must first list some basic components of electric motors including: a "rotor" (the part of the motor which rotates and includes the shaft), a "stator" (which surrounds the rotor but doesn't touch it), "windings" (wire wound into loops around iron cores to produce magnetic fields), a "commutator" (electrical connections which enable current to flow through the windings) and "brushes" which make electrical contact to the commutator sections. (Clearly, "brushless" motors do not have these.) These components can be arranged as shown in Figure 1 for an

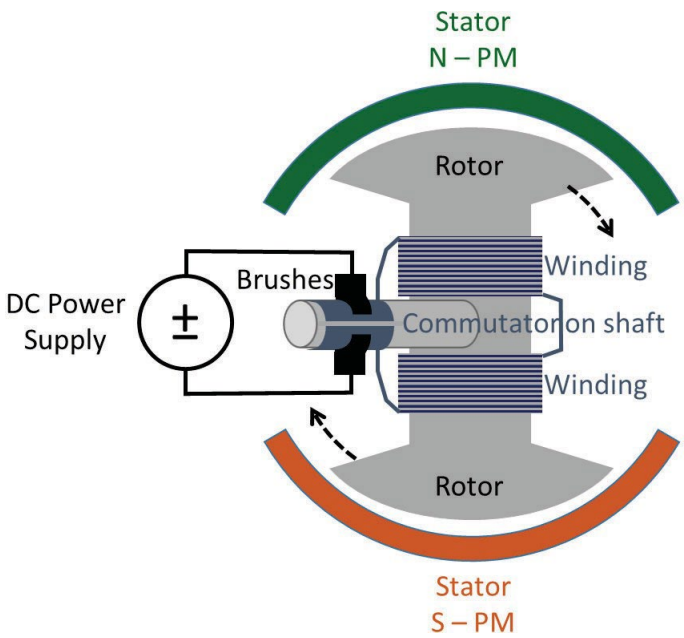

Figure 1. A schematic drawing of an electric motor showing a rotor and shaft, commutator, brushes, stator (as Permanent Magnets (PM)) as well as the windings. The commutator ensures that the current through the windings changes direction to continue to force rotation of the rotor.

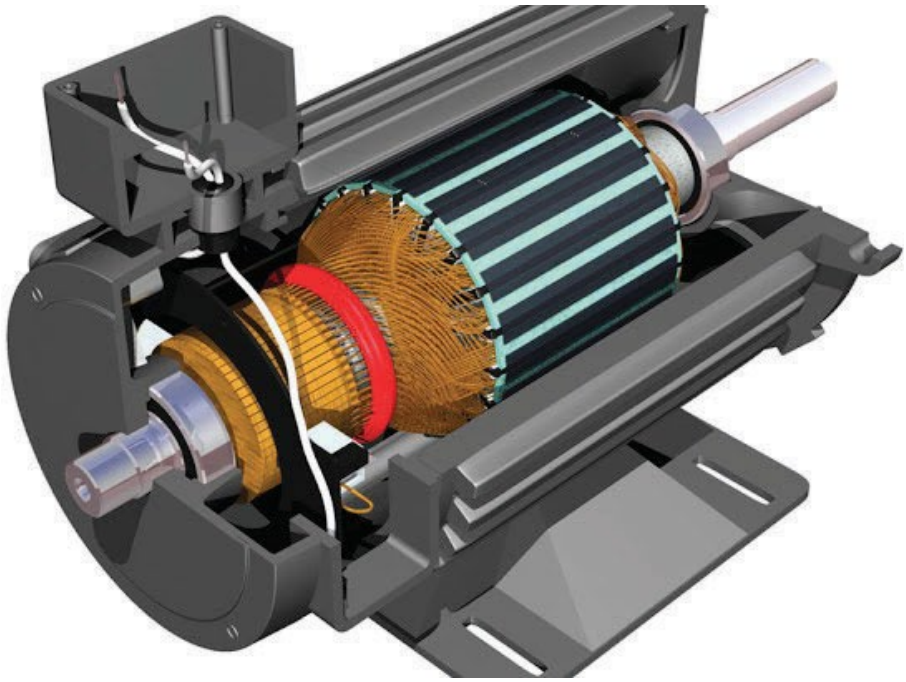

Figure 2. An exploded view of an AC electric motor showing the shaft and rotor, brushes, commutator, windings (in the rotor), stator and outer casing. From http://acmotorkitpicture.blogspot.com/2013/08/ac-motor-exploded-view.html

overly simplified DC motor. The qualitative idea is also reasonably simple. The commutator and brushes ensure that the magnetic field induced on the rotor by the flow of current switchesto the same direction as that produced by the permanent magnets at just the right rotation location of the rotor. As a result, the magnetic north of the rotor faces the north on the stator (N – PM) and south faces south at just the right time. This induces a repulsive force between the rotor and stator. The repulsion pushes the rotor and induces the rotation. Note that the rotational location where the force between stator and rotor reaches a maximum is when the rotor is 90° from the position shown in Figure 1. (The rotor location shown in Figure 1 is actually the minimum force position.) The number of north and south poles on the rotor and stator can be chosen independently and the version shown in Figure 1 is the simplest possible (2+2). One can imagine 4+2, 2+4, 4+4 or 6+4 or any other combination of integers for the numbers of poles. If you cannot imagine these other combinations, then try looking it up in thelibrary or on the internet.

Figure 2 shows an exploded view of an AC electric motor. (There are obviously differences between AC and DC electric motors, but we will not go into those differences here. You can take an elective course which includes this subject if you desire.) Note that the commutator now has many segments (not just the two of Figure 1) and that the rotor has many poles as well. (The rotor poles can be identified as the black lines on the light blue (insulating) background.) The increased number of poles on the rotor and stator make for a more constanttorque as the rotor spins. The 2+2 pole configuration in Figure 1 would have rather large swings in torque as the rotor spins. The swings would go from zero to maximum and cause the rotor speed to increase and decrease too much during each rotation as well! This will be a littlebit of a problem for us in this experiment, actually, because the hobby motors we will use have only a few poles and thus a rather large torque variation during each rotation too.

While it is true that an electric motor converts electrical power to mechanical power; it is also true that motors can be operated in reverse. If you spin the rotor of an electric motor it induces an EMF (electromotive force – voltage) on its windings. This is, in point of fact, using the motor as a “generator” and is what is occurring in windmills and other electric power stations. We will not explore this aspect of electrical machines in the present lab, but may doso in the next lab course.

The focus of this lab will be on a small (hobby scale) DC electric motor. This motor is reasonably typical of a wide range of electric motors and can give you a beginning insight totheir operation and use. An exploded view of this DC electric motor is shown in Figure 3.

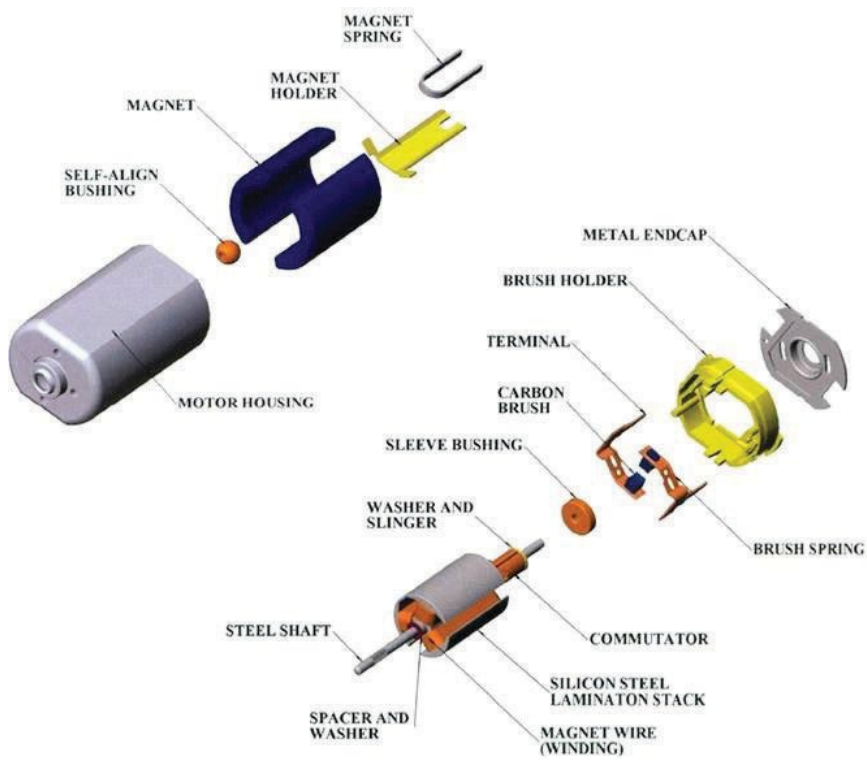

Figure 3. An exploded view of the DC hobby motor to be used in this lab. Note the permanent magnets (stator), coil windings (rotor), brush springs, carbon brushes and commutator. From: http://elect-eng-world1.blogspot.com/2015/07/permanent-magnet-direct-current-pmdc.html

Basic Theory of Operation of DC Electric Motors

Torque vs. Speed Curves and Output Mechanical Power

Direct Current (DC) motors are electro-mechanical machines which operate using direct current electricity and convert electrical power to its mechanical counterpart. The followinglink provides a brief introduction to DC motors which you must watch:

https://youtu.be/LAtPHANEfQo

In this lab, we will examine the electro-mechanical properties of a DC motor called a "hobby" or "toy" motor. We will use a power supply and various masses to test its electrical and mechanical properties. In general, DC motors are able to produce a torque that is reasonably proportional to the angular velocity of the motor. Note that the torque, r, has the units of N*m or force * length. The torque acting on a rod of length $|\bar{R}|$, can be defined as:

$$r = \bar{R} \cdot \bar{F} = |\bar{R}||\bar{F}| \sin(\varphi) \qquad (1)$$

The axis of the rod's rotation sits at the origin, O. Figure 4 schematically illustrates the envisioned rod (thick black line), force (dashed blue line and arrow) and resulting torque. Note

that the torque is defined to be the portion of the force which is perpendicular to the rod and therefore acts to rotate the rod around the origin.

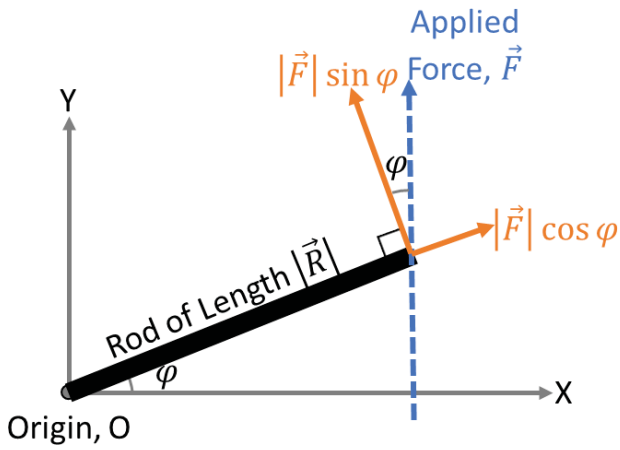

Figure 4. A diagram of force acting on a rod (or rotor) rotating around the axis origin, O, and producing a torque as in equation (1).

A DC motor generates an electrically—induced force to rotate a rod or rotor about its axis. For most electric motors, there exists an approximately linear relationship between the torque, Y, and the angular speed (or angular frequency) ω, of the rotor. (Note that the units of ω are radians/sec.) That linear relationship can be mathematically modelled as:

$$r = (1 - \frac{\omega}{\omega_n})\, r_s \qquad (2)$$

Here: r_s is called the "stall torque" or the torque required to stop the rotor from spinning ($\omega = 0$). ω_n is called the "no-load rotational speed" ($\omega_n = \omega(r = 0)$) or no-load rotational frequency. This is how fast the motor can spin if it is not having to move anything else. Figure 5 is a plot of equation (2). One can see that the amount of torque which the motor can apply decreases as its rotational speed increases. It has the largest torque when stationary and its top rotational speed is ω_n. Like any model of a physical system, equation (2) has limitations.

Real electric motors can be more complicated and require more terms in the torque-speed equation. Those additional terms enable one to model the non-linear aspects of the torque-speed relationship. We will not worry about non-linearity here, but you can learn about it inother courses and various textbooks. We will keep the theory simple and linear for this experiment so that you learn the highest impact aspects of electric motors. You will make measurements in the lab that directly relate to equation (2) as part of characterizing a DC hobby-motor.

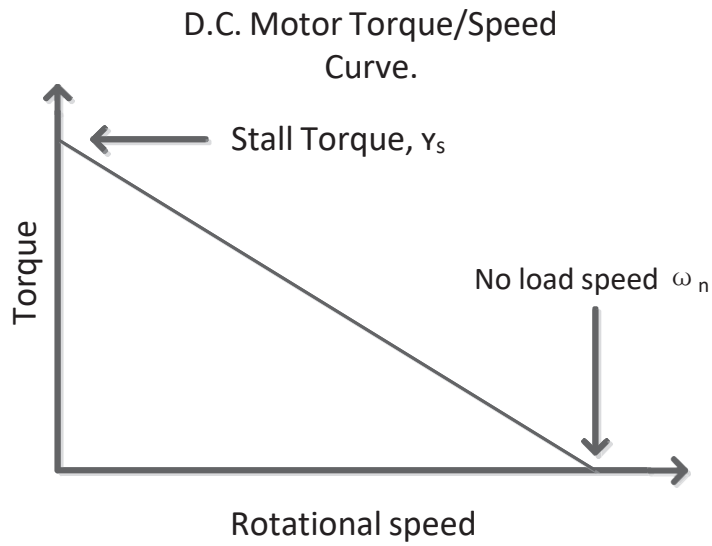

Figure 5. A graphical representation of equation (2) showing the simple linear relationship between the torque on the rotor of an electric motor and its rotational speed (or frequency).

Electric motors obviously have a maximum mechanical power output. They can output onlyso much power and no more. This maximum power can be determined for each motor and onecan use it to describe the motor as well. The input electrical power and the output mechanical power for an electric motor can be written as:

$$Input\ Electrical\ Power = P_{in} = I * V \quad (3)$$

$$Output\ Mechanical\ Power = P_{out} = r * \mathrm{m} \quad (4)$$

The units of power are Nm/s = torque/s = J/s = Watts. Equation (2) specifies a relationship between torque and rotational speed which can be used in equation (4). The result is:

$$P_{out} = (1 - \frac{\omega}{\omega_n})\, r_s * \mathrm{m} = r_s\omega - (r_s/\omega_n)\omega^2 \quad (5)$$

The mechanical power has a parabolic relationship to the rotational speed and can be visualizedas the area of a rectangle ($r * \mathrm{m}$) under the torque-rotational speed curve at each rotational speed. Figures 6 and 7 show some examples of those mechanical power rectangles and help one to visualize why low speeds and high speeds are both lower mechanical power situations.

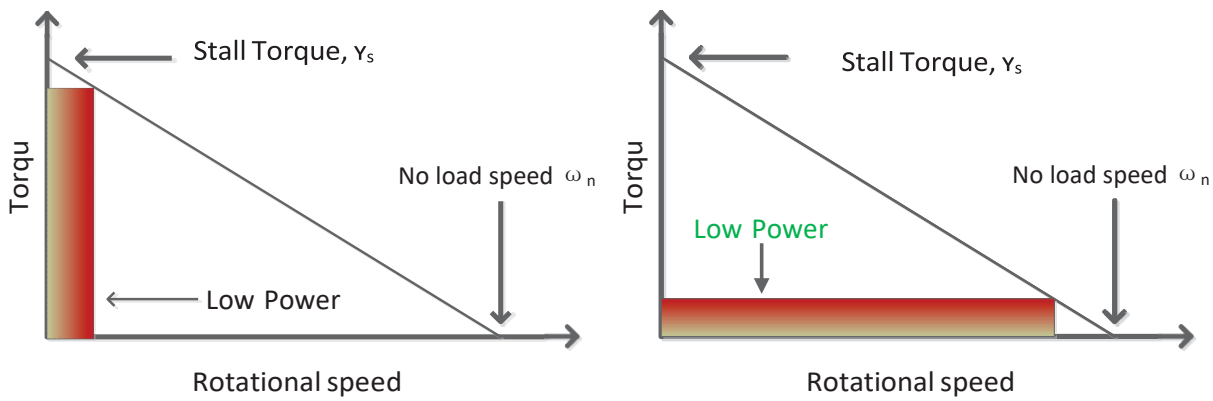

Figure 6. Low mechanical power operating points of an electric motor illustrated on the torque-rotational speed graph for the motor. Power is given by equation (3) $P = r\omega$. Left: Low power output due to low rotational speed (ω). Right: Low power output due to low torque (γ).

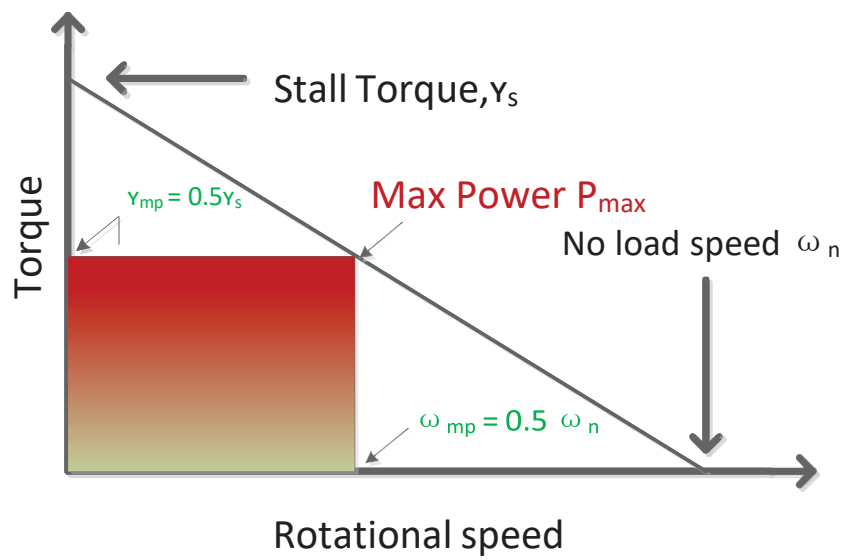

Figure 7. The point of maximum mechanical power output. The area of this rectangle is the largest possible under the torque-rotational speed curve. The torque is half of the stall torque and the rotational speed is half of the no load rotational speed.

Figure 6 and Figure *7* depict operating points of a DC motor. Figure 6 depicts low mechanical output power points while Figure 7 depicts the maximum output power operatingpoint for the motor. This maximum power point can be found analytically by differentiating equation (5) with respect to the rotational speed and setting it equal to zero. (This is a maximum because the second derivative is always less than zero.) That is:

$$\frac{6P}{6\omega} = (1 - \frac{\omega}{\omega_n})\, r_s - \frac{\omega}{\omega_n} r_s \;\Xi\; 0$$

$$\rightarrow \omega = \frac{\omega n}{2} \;\Xi\; \omega_{mp} \qquad (6)$$

Here: ω_{mp} is defined to be the rotational speed corresponding to the maximum output power. Using this value in equation (4) or (5) gives one that the torque at the maximum power point is $\gamma_s/2$ and the maximum mechanical power is

$$P_{max} = ({}^{\omega_n r_s}/_4) \quad (7)$$

This rotational speed at which the motor is outputting its maximum mechanical power is adesirable point at which to operate a DC motor; however, it does not have to correspond to the point of maximum electrical power input to the motor! It is, therefore, important to find the relationship between the electrical input power to the motor and the mechanical power outputby the motor as part of establishing the electro-mechanical efficiency of the motor. One might want to operate the motor at the highest efficiency point for an electric vehicle (for instance) inorder to drive the longest possible distance between recharges and attain the lowest operating costs. While efficiency is great, maximum efficiency and maximum acceleration (torque) can beat two very different operating points. The motor efficiency can be calculated as a ratio of equations (4) and (3) = P_{out}/P_{in}. It can be found experimentally and you should do so in this lab.

Finding the Torque vs. Speed Curve in Specifications and in the Lab

Any motor you purchase ought to come with a set of specifications detailing its basic operation and parameters. The hobby motors used in this lab are no different. We will makemeasurements of some of those basic parameters in order to obtain a better understanding than the (rather minimal) specifications provided with the motors. Table I. lists those motor specifications.

Table I. Technical Specifications of the P711A Hobby Motor (DC Electric Motor)

P711A Hobby-Motor Technical Specifications			
Operating Temperature	-10	+ 60	Degrees Centigrade
Rated Voltage	6.0		Vdc
Rated Load	10.0		g-cm
No-Load Current	70		mA (Max)
No-Load Speed	9100	±1800	RPM
Loaded Current	250		mA (Max)
Loaded Speed	4500	±1500	RPM
Starting Torque	20.0		g-cm
Starting Voltage	2.0		V
Stall Current	500		mA (Max)
Body Size	27.5 mm x 20 mm x 15 mm		
Shaft Size	8 mm (Length) x 2 mm (Diameter)		
Weight	17.5		grams (g)
Operating Voltage Range (Typical)	4.5	9.0	Vdc

The meaning of some of the specifications are obvious such as the operating temperatureand dimensional values. Other specifications might not be as clear and so we will go through them briefly here. The “rated voltage” and “rated load” describe typical operating conditions for the motor. One can operate the motor under conditions different than these, but these

conditions demonstrate its capabilities. The next two items, the "No-Load Current" and "No-Load Speed" describe the amount of current flowing into the motor and the rotational speed of the motor shaft when there is nothing attached to the shaft. The applied Voltage for the "No-Load" case need not be separately specified because it is 6.0 V (the "Rated Voltage"). The shaft spins at 9100 RPM ±1800 RPM with No-Load, which ought to be close to the maximum rotational speed for the motor (at the Rated Voltage). Thus the table gives one an estimate for ω_n in equation (5). The "Loaded Current" and "Loaded Speed" describe the motor operating conditions when the "Rated Load" of 10 g-cm torque is put on the motor shaft. The load torque causes the rotational speed of the motor shaft to decrease and the current increases as the electrical power needed by the motor to rotate against the torque increases too. One can see that the "Loaded Speed" is about half of the No-Load speed (which we just said was an approximation for ω_n) and therefore, is reasonably close to the maximum mechanical power operating point of the motor. The "Starting Torque" describes the "torque" that the motor can produce when it is just beginning to rotate. This is an estimate of the "Stall Torque" (r_s) in equation (5) using non-standard units. The units of "g-cm" can easily be converted to kg-m but the units of torque are N-m. The difference is a unit of acceleration (m/s^2). Thus one uses the value of gravity (9.8 m/s^2) to convert g-cm to N-m. 20 g-cm becomes ~0.002 N-m. Thus the table gives estimates of both r_s and ω_n and allows one to draw the torque speed curve at the Rated Voltage operating condition. Using equation (7) one can find that the maximum mechanical power for the motor is ~470 mW (at the Rated Voltage). Note that the loaded current is 250 mA (max) and the rated voltage is 6 V so the electrical input power is 1.5 W and the efficiency is ~31% The "Starting Voltage" describes the minimum voltage required to start the motor shaft spinning under "No-Load" conditions. Less than 2 V applied to the motor and the shaft will remain motionless even though electrical power is being deposited. Finally, the "Stall Current" describes the typical amount of current which flows through the motor windingswhen the shaft is held motionless and the Rated Voltage is applied. Note that 500 mA at 6 Volts corresponds to 3 Watts and could easily result in the motor overheating and burning up if applied for too long.

The primary objective in this lab will be to measure the torque-speed curve for these hobby motors at two operating voltages other than the Rated Voltage of 6 V. You will need to worry about whether or not the voltages you choose cause the motor to rotate too fast for your measurements! In addition, you will analyze the curves to determine the maximum mechanical power and power efficiency of the motor. To measure the torque-speed curve will require that you fabricate at least 8 different loads (masses) to present at least 8 different torques on the hobby motor's shaft. You will need to use a stopwatch (a cell phone will do) and a tape measure too.

Example Problem: Loading a DC motor with different masses (torques):

Rotating DC motors are characterized by their Torque vs. Speed curves as noted above. Those curves can be measured using sophisticated equipment to induced calibrated torques on the motor shaft to go along with precise measurements of the rotational speeds. Instead of this, we must use a rather primitive method of making the same measurements in this lab experience. We will hang a weight on the shaft of the motor by a string and use the motor to

wind it up. Measuring the time it takes to rise a known distance will allow us to estimate the speed at a known torque.

Imagine that a DC electric motor has a spindle attached to its shaft as an extension. The spindle is 7 mm in diameter and has a disk at the end. A drawing can be seen in Figure 8. A string is attached to the spindle and winds onto it as the motor spins. A mass is attached at thebottom of the string and applies a downward force on the spindle equal to the mass of the weight * the acceleration of gravity, m*g. This downward force results in a torque on the spindle which the motor must overcome to wind up the string as shown on the right hand side of Figure 8.

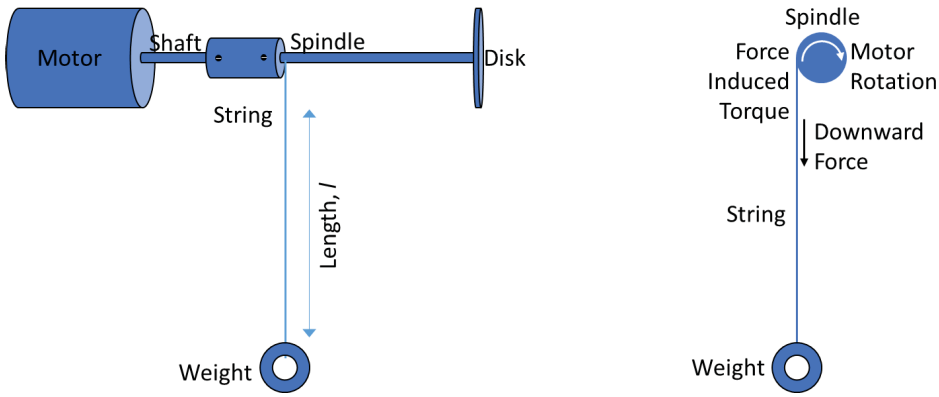

Figure 8. Left: A DC motor with a spindle attached and a string with a weight to be wound upon that spindle. Right: Looking along the axis of the spindle. One can see how the weight induces a downward force and torque on the spindle. The motor rotates clockwise to pull upthe weight.

When the weight is changed, the torque on the spindle (and motor shaft) changes as well and causes the rotational speed of the motor to change too. One can measure the motor rotational speed for various weights (torque levels) and thereby recover the torque vs. speed curve for the motor. A linear fit of this curve allows one to find the intercepts with the torque axis (stall torque) and the speed axis (no-load speed). One can thereby characterize the motor. The measurements are made by measuring the time (dt) it takes to raise the weight a set length(*l*). Thus the average speed is given by v = *l*/dt in m/s. The spindle winds up a length of string = 2uR each rotation so that the rotational speed of the spindle (and motor) must be v/(2uR) rotations per second (RPS). The torque is simply calculated according to equation (1) as mgR. Thus one can obtain several values of the torque speed curve by making measurements using several masses.

As an example problem: assume that the measurements in Table II were obtained with 96 Volts applied to a DC motor that is used on a Tesla vehicle. (Tesla would not make such measurements in the same way that we will in lab! Can you figure out why?) The voltage waskept constant and the DC current flowing through the motor was measured to allow one to determine the input electrical power. The angular speed was measured in “rotations per minute” (RPM). (Note: 60 RPM = 1 RPS and corresponds to a rotation frequency of 1 Hz or ω =2u Radians/s.)

Table II. Operating data for a fictitious DC electric motor. The data has been generated to be similar to what one might expect for a DC motor on an electric vehicle like a Tesla.

Torque (g-cm)	Rotation Speed (RPM)	Current (mA)
50,000	3,356	65,625
150,000	2,868	96,875
250,000	2,441	128,125
350,000	2,022	159,375
450,000	1,505	190,625
550,000	1,071	221,875
650,000	673	253,125

Problem: Use Table *II* to accomplish the following:

1. Plot a torque vs. speed curve for this motor.
2. Fit the torque vs. speed curve to a line and use the fit equation to calculate the
    a. Stall Torque,
    b. No-Load Rotational Speed,
    c. Maximum Mechanical Power output for the motor.
3. Calculate and plot:
    a. the mechanical output power as a function of the rotation speed, and
    b. the electrical input power as a function of the rotation speed.

Example Problem Solution:

Plotting the torque speed curve for the data in Table II is simple and one can use a least squares method to fit a line through the data. Excel allows one to do this using the "trendline" function and Matlab allows one to do this using the "lsqcurvefit" function. (x = lsqcurvefit (fun,x0,xdata,ydata)) The plot and least squares fit are shown in Figure 9. The linearfit equation (also given in Figure 9) is

$$\text{Torque} = \text{-}222.41*(\text{Rotational Speed}) + 792{,}776 \qquad (8)$$

Equation (8) makes it simple to find the Stall Torque, No-Load Rotational Speed and Maximum Mechanical Power output for the motor. The stall torque is the torque at which the rotational frequency = 0. Thus r_s = 792,776 g-cm = 7.9 kg-m and corresponds to 77.7 N-m. (The conversion from kg-m to N-m requires multiplying by the acceleration due to gravity = 9.8 m/s^2.) The No-Load rotational speed, ω_n, is the speed at which the motor spins when the torque = 0. Thus, ω_n = 792,776 / 222.41 = 3564 RPM = 59.4 RPS and corresponds to 373.3 Radians/s (in MKS units). The maximum mechanical output power is given in equation (7) butmust be calculated using MKS units for both ω_n and r_s. The result is 7,250 Watts.

It is simple to calculate the electrical power from the data in Table II. It is just the voltage * current (96 V * I). Calculating the mechanical power is also easy. It is the rotational speed * torque (in MKS units). These are both plotted in Figure 10. Note that the power efficiency curve (mechanical power / electrical power) will not peak at the same rotational speed as the

maximum in the mechanical power for this motor.

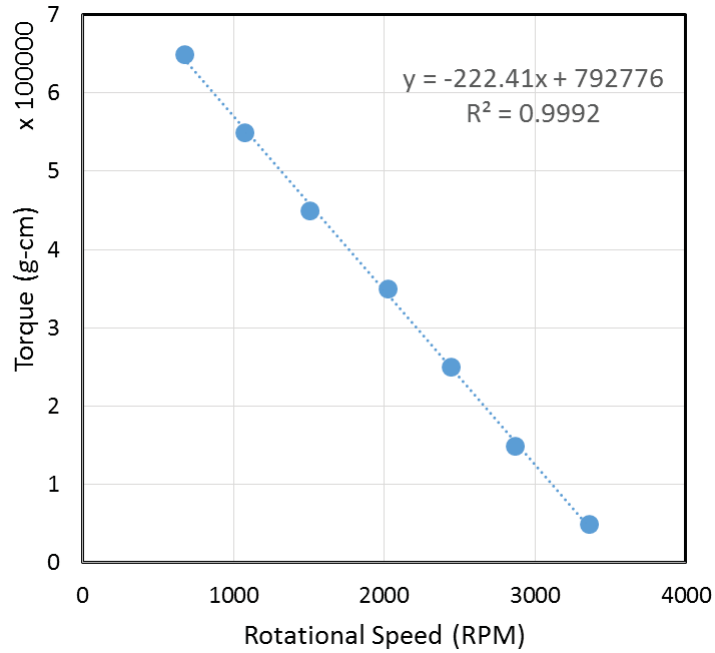

Figure 9. The Torque vs. Speed curve for the data of Table II. The torque values range from 0 to 700,000 g-cm. The linear fit equation is shown inside the plot.

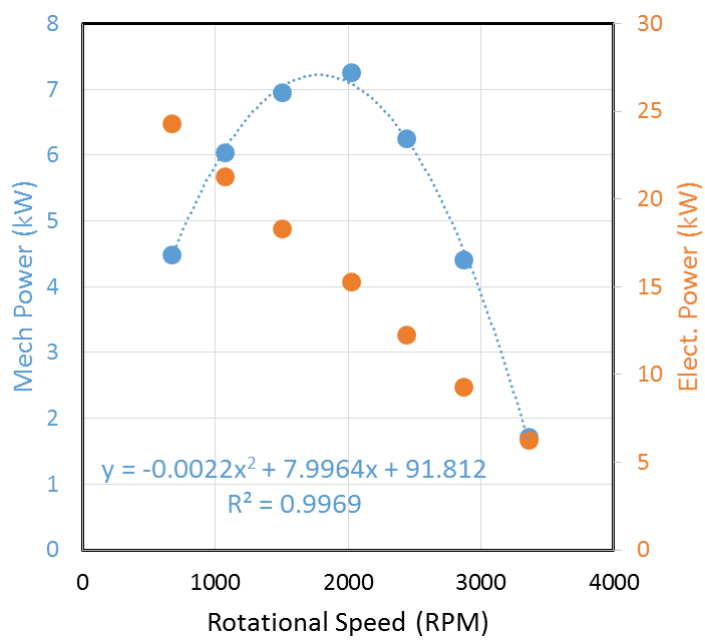

Figure 10. Mechanical Output Power (blue) and Electrical Input Power (orange) as a function of Motor Rotational Speed. The quadratic fit is to the mechanical power and shows how it has a maximum near $\omega_n/2$.

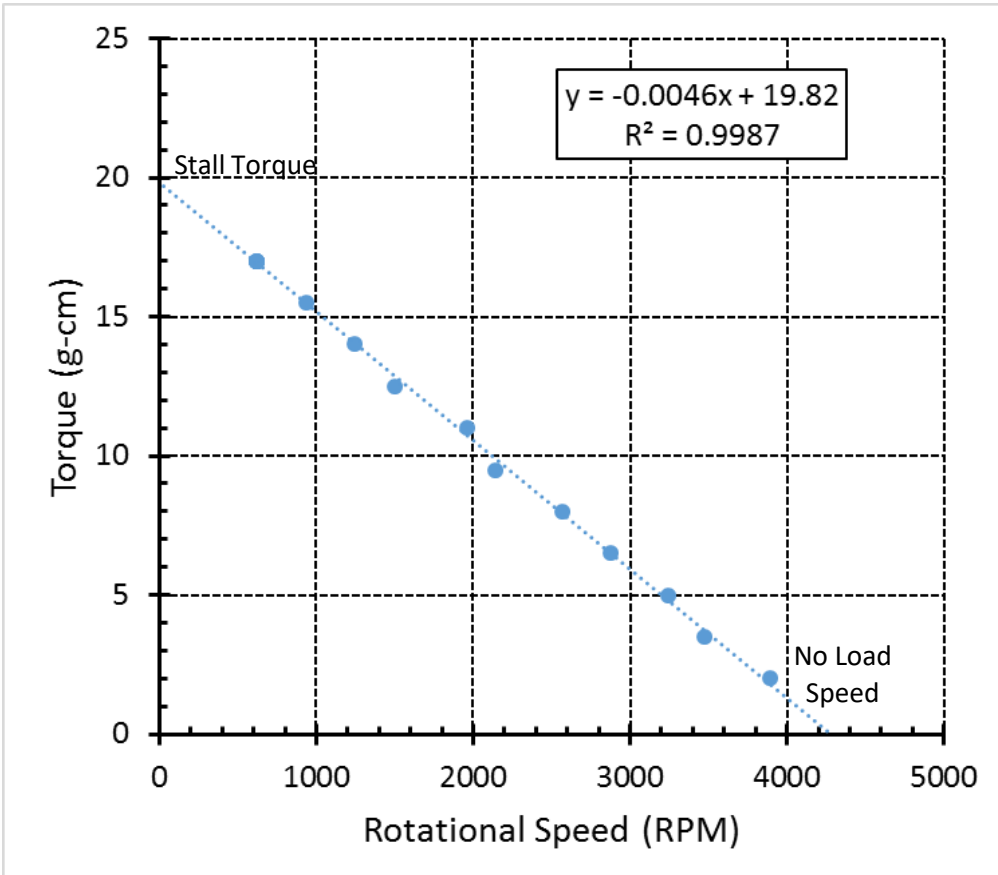

Figure 2.1: An example of a torque vs. rotational speed curve for a hobby motor. The linear fit to the data points gives a stall torque of 19.82 g-cmand a no-load rotational speed of 4308 RPM (=19.82/0.0046).

- *For more details on DC Electric motors refer to the other document which includes theory of operation of the motors*

Materials required:

1. Standard testing station
2. P711A hobby-motor
3. 2 mm spindle
4. 1.5 m string
5. P711A hobby-motor stand
6. 1 m ruler (or measuring tape)
7. Tachometer
8. Plastic wheel with all but one spoke blackened or tape on its side.

3. Pre Lab

1. Values of torque are given in a variety of units. For example: You saw torques specified in units of gram*cm (g-cm) in Table I. Convert the following non-MKS torque values to torques in the MKS units of N-m. (Hint: acceleration due to gravity is 9.8 m/s2.)
a. 57 g-cm
b. 38 kg-m
c. 1.7 J
2. You have a motor with a shaft diameter of 25 mm from which you have hung a weight of 1.1 kg on a thin wire. (Assume that the mass of the wire mass is very small compared to that of the weight.) What is the torque on the motor's shaft?
a. In g-cm?
b. In kg-m?
c. In N-m?
3. You turn on the motor in problem 2 and it pulls the weight up 79 cm in 2.5 seconds.
a. What is the total amount of mechanical energy expended by the motor in pulling up the weight? (Give your answer in Joules.)
b. What is the average rotational speed of the motor shaft while pulling up the weight?
c. What is the average mechanical power output by the motor in Watts?
4. You are given the following data for a DC hobby-motor with an operating voltage of 6 Volts.

Table III

Torque (g-cm)	Rotation Speed (RPM)	Current (mA)
10	4413	330
25	3684	375
40	3147	420
55	2216	465
70	1499	510
85	768	555
90	482	570

a. Plot the Torque vs. Speed curve for this motor.
b. Fit the Torque versus Speed curve and use the fit to find
i. The stall torque, τs,
ii. The No-load speed, ωn, and
iii. The maximum mechanical power for the motor.
c. Plot the power efficiency vs. rotational speed curve for the motor. What is the highest efficiency that this motor attains?
5. You are given a DC motor with the following specifications:

Fictitious Motor Technical Specifications			
Operating Temperature	-10	+ 60	Degrees Centigrade
Rated Voltage	50.0		Vdc
Rated Load	100.0		kg-m
No-Load Current			A (Max)
No-Load Speed			A (Max)
Loaded Speed	100	±7	RPM
Starting Torque	20.0		kg-m
Starting Voltage	2.0		V
Stall Current	500		A (Max)

a. What is the maximum mechanical power for this motor? (Give your answer in Watts.)
b. Assuming the motor is 30% efficient and runs at the rated voltage, how much. current must flow through the motor at maximum output power.
c. What is the fastest acceleration possible for a 2000 lb. (907 kg) automobile using 2 of these motors as the primary drive force? Assume the wheel radius to be 15 inches.

4.Procedure

1. Initial Lab Prep:
   a. Obtain a P711A hobby-motor and a plastic wheel.
      i. Make sure that all but one of the spokes of the plastic wheel are non-reflecting.
      ii. Securely connect the wheel to the shaft and make sure that it is not tilted.
   b. Examine the DC motor.
      i. Identify its positive and negative terminals.
      ii. Re-read the motor's data sheet carefully and pay special attention to its no-load rotational speed at the rated voltage. In the first part of today's experiment, wewill be measuring this (and more).
   c. Mount the motor onto the stand securely, but without squeezing the motor too tightly. Squeezing the motor too tightly will add friction on the rotor and can cause your measurements to be incorrect.
   d. Turn on the DC power supply and make sure that the +25 V output is set to source 0 V and **is not yet enabled,** voltage to the motor should not surpass 9V**.**
      i. Connect the DC power supply, DC voltmeter and DC ammeter to the terminals of the motor.
      ii. When all is connected correctly, you should have what is shown in Fig. 3.1.

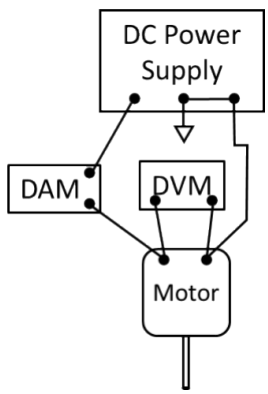

Figure 3.1: Electrical Connections for the first tests. The Digital Ammeter (DAM – Keysight 34401) is connected in series with the motor and the Digital Voltmeter (DVM – Keysight 34401) is connected in parallel. The 25 V output of the DC power supply is connected in parallel too.

2. Familiarize yourself with the Tachometer.
   a. Since it uses a laser, do not operate it until your TA has provided instruction on the safe operation of this device. ALWAYS ensure that the laser beam is aimed at a nearby wall and never into the center of the room or at a person. It must always be directed at a safe target as you make your measurements.
   b. Place the tachometer on a solid surface pointing at the wheel. The red beam should only be lighting the tape placed on the side of the wheel. Turn on the motor by applying2 V from the power supply and test that the reading is stable. At 2 V, the tachometer reading should be roughly 2500 ± 300 RPM. Proceed to the next step when you are able to make stable readings using the tachometer.

3. Measuring the "No-Load Speed" for the motor.
   a. Set the power supply to 2 V and use the tachometer to measure the approximate no-load speed of the motor. Then record:
      i. The voltage across the motor leads (DVM output but ~2V in this case!),
      ii. The current flowing into the motor (DAM output) and
      iii. The motor rotational speed (Tachometer output, when the value is ~stable).
   b. Repeat 3.a for at least 5 different input voltages between 2 and 9 Volts (2, 4, 5, 7, and 9).
   c. When finished, you should have a table of data with four columns: Voltage (V), Current (A) as well as Rotational Speed (RPM). Add three more columns to the table:
      i. Electrical Power (Watts) which is simply = Voltage * Current, and
      ii. Rotational Speed in Rotations / second (RPS) which is simply RPM/60.
      iii. Rotational Speed in Radians/second. This is the "Angular Frequency" = ω. It is calculated by noting that each rotation encompasses 2π radians. ω = RPS * 2π.
   d. Make a quick plot of your data. Plot Max-RPM versus Voltage and do a fit to the data in either Matlab (preferred) or MS Excel. Show your data and your fit to your TA for approval.

4. Measuring the torque vs rotational speed curves for the motor.
   a. Attach a string on the shaft as demonstrated by your TA and shown in Fig. 3.2. Attach awheel at the end of the spindle.

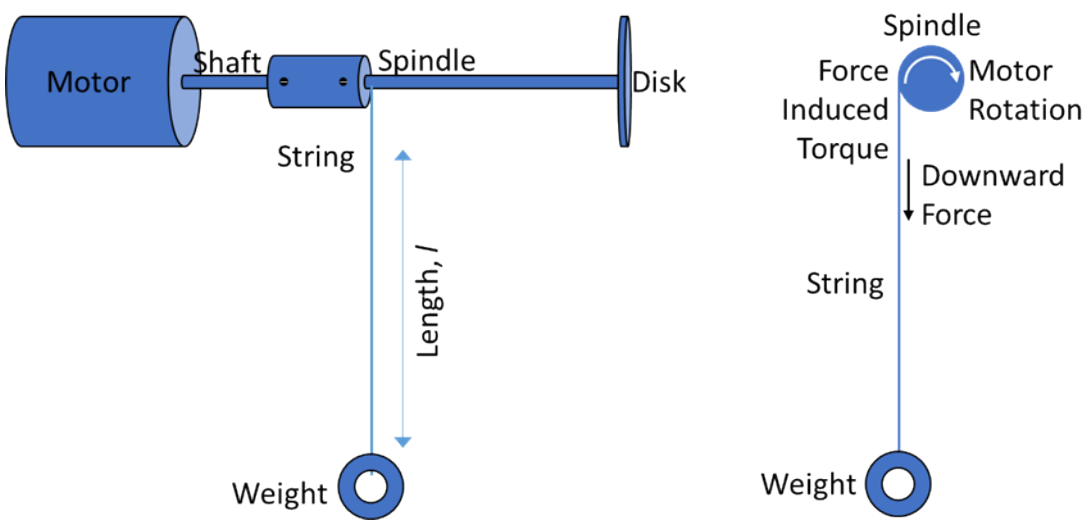

Figure 3.2: The motor with attached spindle and string to wind up a weight. (Left) Schematic diagram of the setup showing the spindle attached to the motor shaft, the string, and the weight. (Right) Diagram of the force induced on the motor spindle by the weight.

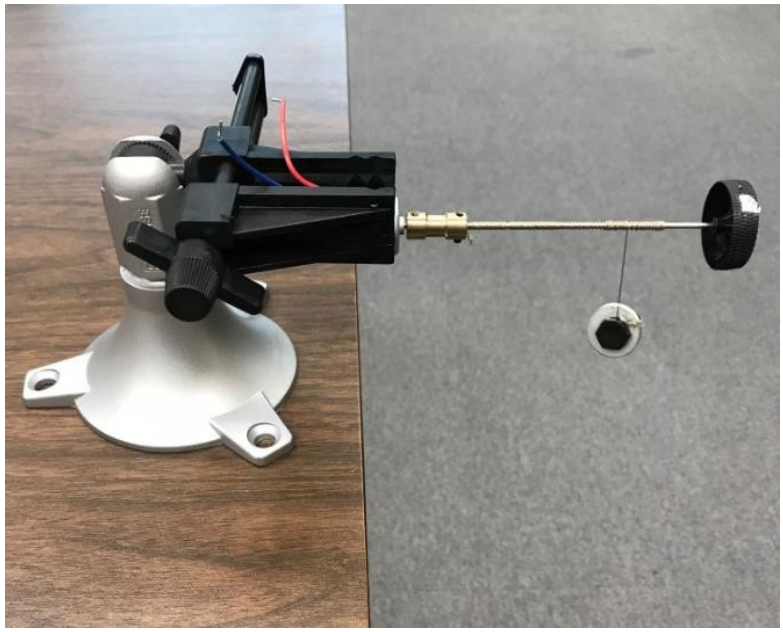

Figure 3.3: Experimental Set-up

i. We are going to attach weights (washers) at the end of the string and measure the rise time of the weights from the ground to the level of your lab table. Each washer is approximately 5.2 grams.
ii. Attach 5g weights at the end of approximately 1m of string, Turn on the DC power supply. (Make sure that it is at your chosen voltage setting.)
iii. With your cell phone stopwatch, measure the rise time of the weights from the ground level to your table level.
iv. Record the Voltage, the rise time and the current readings.
v. Add or remove weights from the end of the string and repeat the experiment, recording the voltage, rise time and the ammeter readings each time.
Be sure to put the weights on in a randomized order! *Do not simply add weightseach time. Instead, choose the weight to use in a randomized order. This helps to prevent systematic errors in the experiment. Systematic errors include things like the motor heating up (and performing differently as a result).*
vi. To increase the accuracy of the experiment, repeat the experiment with each weight at least 2 times and preferably 3 times. Obtain the average values for both the rise time and current reading from the ammeter.
vii. Enter your data into a table similar to Table I, shown below.

b. Repeat your measurements for both of your chosen voltage settings.

5. Data Analysis.
   a. Use your measurements to calculate and plot:
      i. The motor torque versus rotational speed curve at each voltage setting. Be sure tofit the curve and find both the motor stall torque and the motor no-load rotationalspeed at both voltages.
      ii. The electrical power entering the motor as a function of the rotational speed at

each voltage,

iii. The mechanical power exerted by the motor as a function of the rotational speedat each voltage,

iv. The motor power efficiency (= mechanical power / electrical power) as a function of the rotational speed at each voltage,

v. The torque versus motor current curve at each voltage.

b. Comment on each of the graphs you have produced.

i. The electrical power: Try to explain the behavior found in this graph. Why should the electrical power entering the motor depend upon the voltage and rotational speed?

ii. The mechanical power: Try to explain the behavior seen in this graph too. Why should the mechanical power of the motor depend upon the voltage and rotational speed?

iii. The motor power efficiency: What settings would you use to produce mechanical power as efficiently as possible? Why should the efficiency vary with Voltage and rotational speed?

iv. The torque versus motor current curve at each voltage: Why should the torque produced by the motor depend on the current flowing through it?

c. Write up your report.

Table I. An example table for entering the torque versus rotational speed data for part 5.a of the laboratory. Notes: (1) The data should be obtained in a random order and not by simply increasing the weight each time. (2) You must fill out a table like this one for two (2) different voltage settings.

Distance (cm)	Voltage (V)	Mass (g)	Time (s)	Current (mA)	Speed (m/s)	Angular velocity (rad/s)	Angular velocity (rpm)
85 cm		0 g					
		5					
		10					
		20					
		30					
		40					
		50 g					

Important Points:

- Calculate/note the data in the form of correct units already mentioned.
- Note that all the required equations related to the calculations in the lab are already provided in the DC Electric motors material.
- When plotting graphs choose scatter plot and draw a chain line in MS Excel.